

Python

```
main.py x +
1 def adicionar(item, lista=[]):
2     lista.append(item)
3     return lista
4
5 print(adicionar(1))
6 print(adicionar(2))
```

Console I/O AI Agent

```
[1]
[1, 2]
```

Valores padrão em parâmetros mutáveis

Java

```
Main.java x +
1 static void zera(int[] v, int n) {
2     v[0] = 0;
3     n = 0;
4 }
5 int[] v = {5, 5}; int n = 5;
6 zera(v, n);
7 System.out.println(v[0] + " " + n);
8
```

Console I/O AI Agent Exit 1

```
Main.java:6: error: invalid method declaration; return type required
zera(v, n);
^
Main.java:6: error: <identifier> expected
zera(v, n);
^
Main.java:6: error: <identifier> expected
zera(v, n);
^
Main.java:7: error: class, interface, annotation type, enum, record, method
or field expected
System.out.println(v[0] + " " + n);
^
4 errors
```

Passagem por valor

Python

```
main.py x +
1 fs = [lambda: i for i in range(3)]
2 print([f() for f in fs])
```

Console I/O AI Agent 136 ms

```
[2, 2, 2]
```

Fechamentos

C

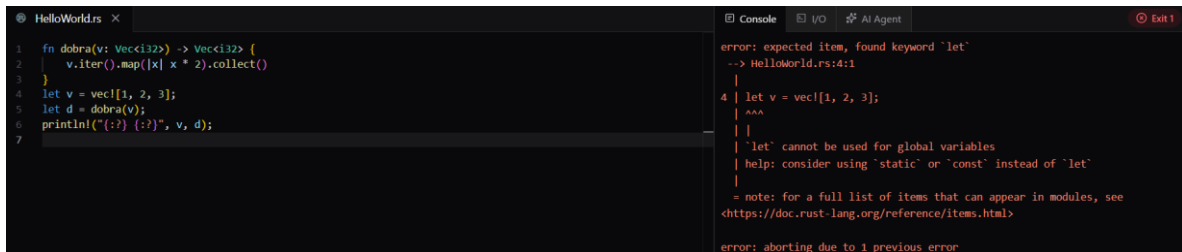
```
C Main.c x +
1 int contador(void) {
2     static int n = 0;
3     return ++n;
4 }
5 // em main:
6 contador(); contador();
7 printf("%d\n", contador());
8
```

Console I/O AI Agent Exit 1

```
Main.c:6:1: warning: data definition has no type or storage class
6 | contador(); contador();
  | ~~~~~
Main.c:6:1: error: type defaults to 'int' in declaration of 'contador' [-Wimplicit-int]
Main.c:6:13: warning: data definition has no type or storage class
6 | contador(); contador();
  | ~~~~~
Main.c:6:13: error: type defaults to 'int' in declaration of 'contador' [-Wimplicit-int]
Main.c:7:8: error: expected declaration specifiers or '...' before string
constant
7 | printf("%d\n", contador());
  | ~~~~~
Main.c:7:16: error: expected declaration specifiers or '...' before
'contador'
7 | printf("%d\n", contador());
  | ~~~~~
```

Variáveis locais estáticas

Rust



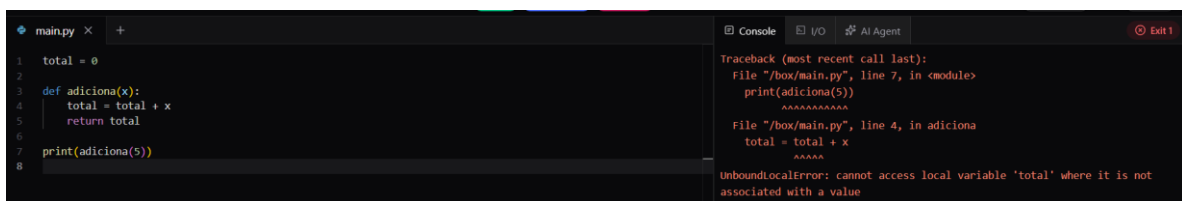
```
fn dobra(v: Vec<i32>) -> Vec<i32> {  
    v.iter().map(|x| x * 2).collect()  
}  
  
let v = vec![1, 2, 3];  
let d = dobra(v);  
println!("{:2} {:?}", v, d);  
}
```

error: expected item, found keyword `let`
--> HelloWorld.rs:4:1
|
4 | let v = vec![1, 2, 3];
| ^^^
| |
| 'let' cannot be used for global variables
| help: consider using `static` or `const` instead of `let`
|
= note: for a full list of items that can appear in modules, see
https://doc.rust-lang.org/reference/items.html

error: aborting due to 1 previous error

Sintaxe de execução em Rust

Python



```
total = 0  
  
def adiciona(x):  
    total = total + x  
    return total  
  
print(adiciona(5))
```

Traceback (most recent call last):
File "/box/main.py", line 7, in <module>
 print(adiciona(5))
 ^^^^^^^^^
File "/box/main.py", line 4, in adiciona
 total = total + x
 ^^^^^
UnboundLocalError: cannot access local variable 'total' where it is not associated with a value

Escopo local vs variáveis do escopo externo